

MATH 8090: An Overview of Time Series Analysis

Readings: CC08 Chapter 1; SS17 Chapter 1.1; BD16 Chapter
1.1–1.3

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Week 1

Agenda

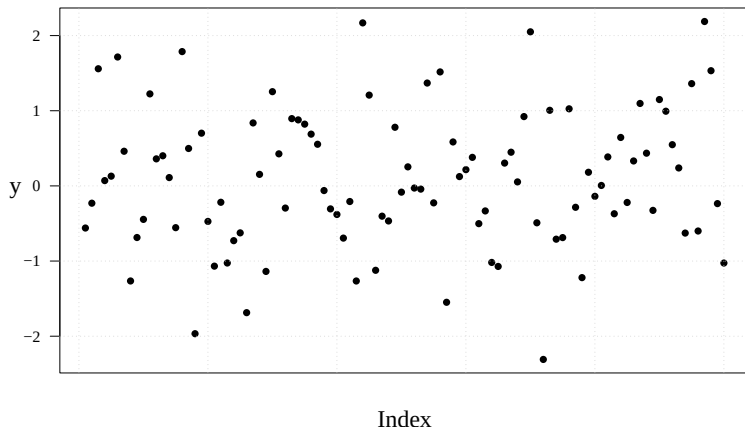
Brief Review of Statistical Inference

Time Series Data

Time Series Models

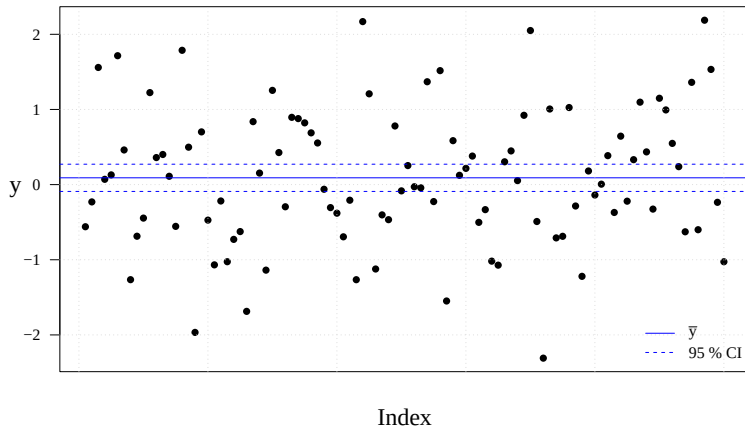
Objectives of Time Series Analysis

Statistical Inference from a Sample



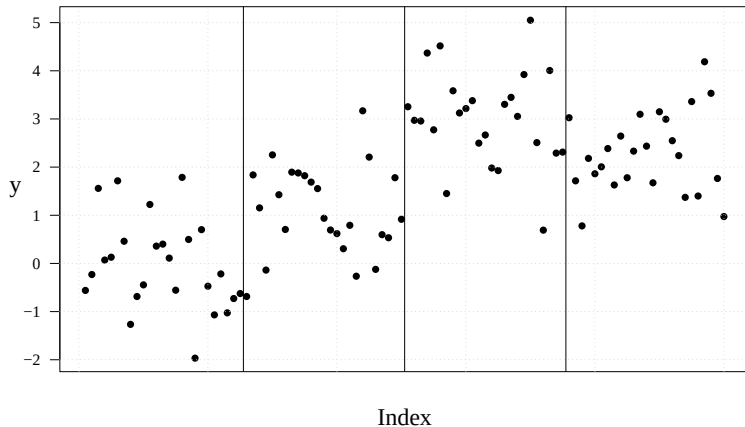
How can we use a random sample to infer the population mean?

Statistical Inference from a Sample



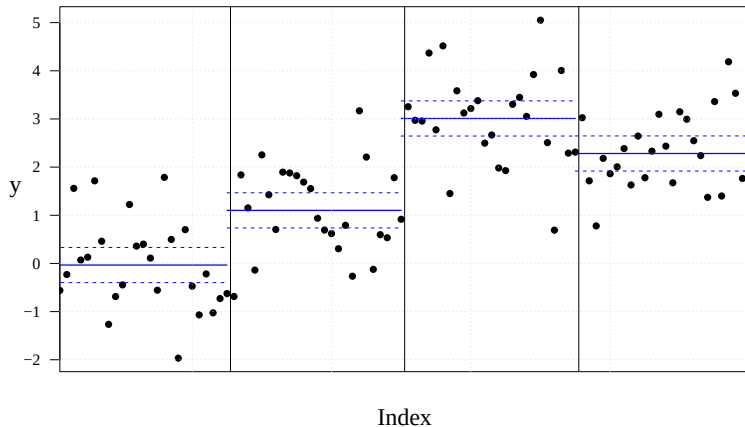
How do we obtain the point and interval estimates?

Statistical Inference for Group Means



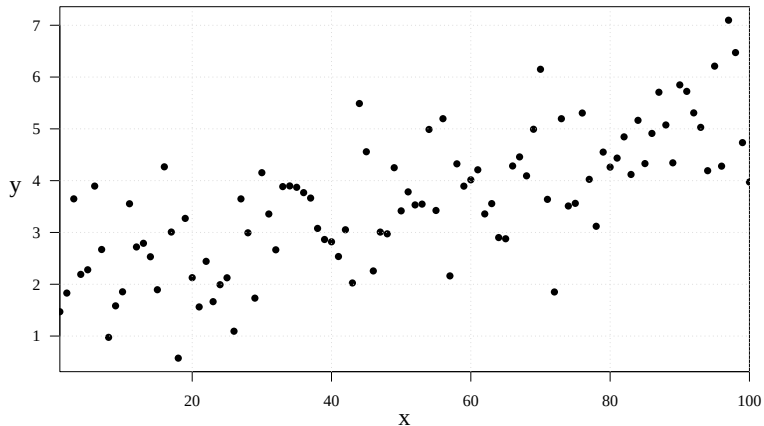
Are the group means significantly different?

Statistical Inference for Group Means



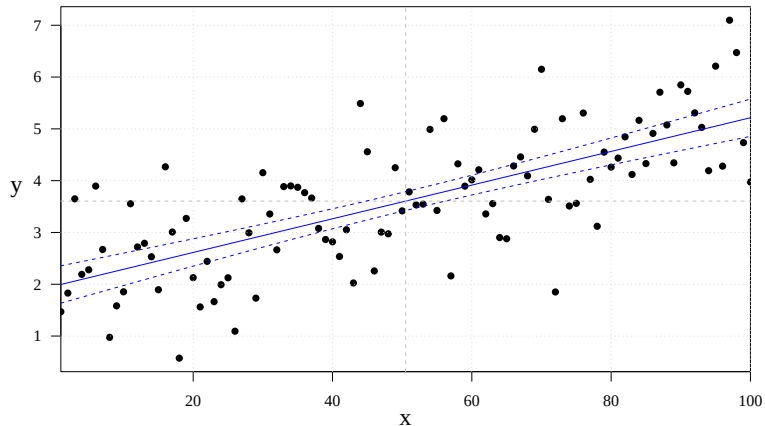
Which statistical technique is used here?

Statistical Inference for Conditional Mean



How does y change conditionally on x ?

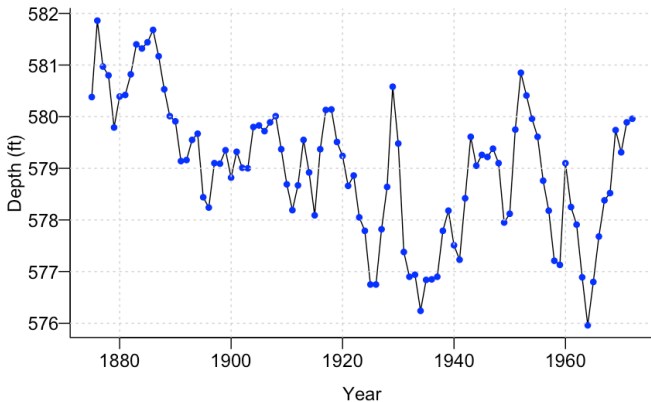
Statistical Inference for Conditional Mean Curve



Which statistical technique is used here?

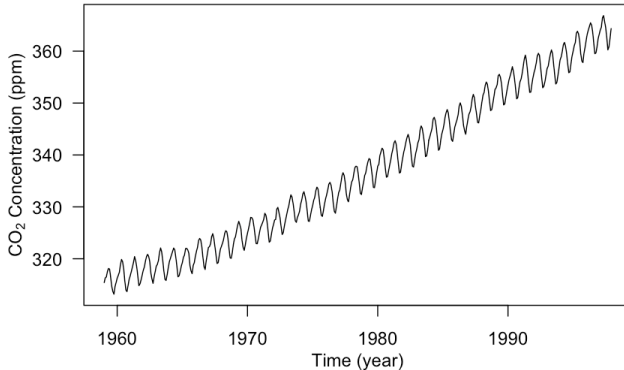
Level of Lake Huron 1875–1972 [Source: Brockwell & Davis, 1991]

```
```{r}
par(mar = c(3.2, 3.2, 0.5, 0.5), mgp = c(2, 0.5, 0), bty = "L")
data(LakeHuron)
plot(LakeHuron, ylab = "Depth (ft)", xlab = "Year", las = 1)
points(LakeHuron, cex = 0.8, col = "blue", pch = 16)
grid()
```
```

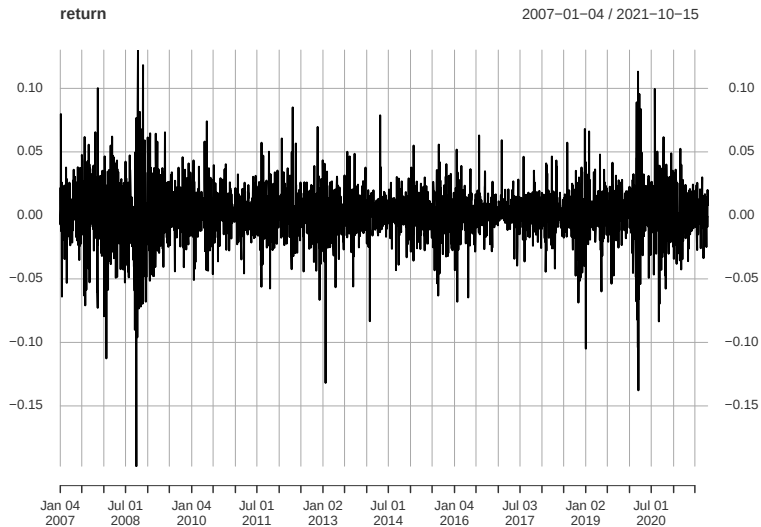


Mauna Loa Atmospheric CO₂ Concentration [Source: Keeling & Whorf, Scripps Institution of Oceanography]

```
####{r}
data(co2)
par(mar = c(3.8, 4, 0.8, 0.6))
plot(co2, las = 1, xlab = "", ylab = "")
mtext("Time (year)", side = 1, line = 2)
mtext(expression(paste("CO"[2], " Concentration (ppm)")), side = 2, line = 2.5)
####
```

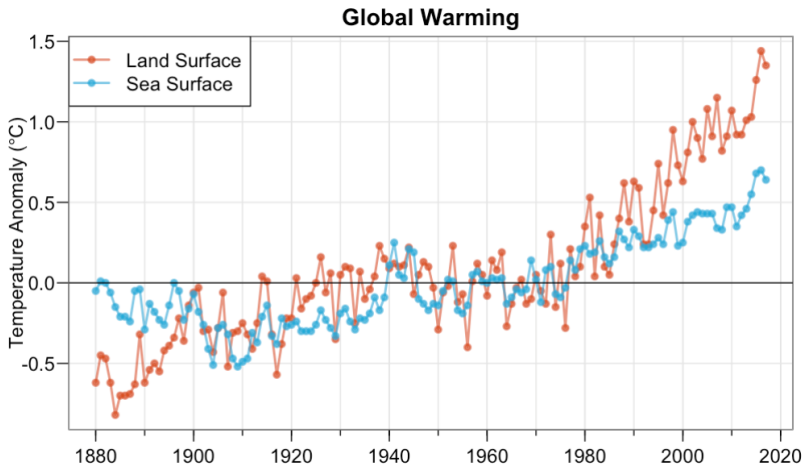


Apple Stock Log Returns



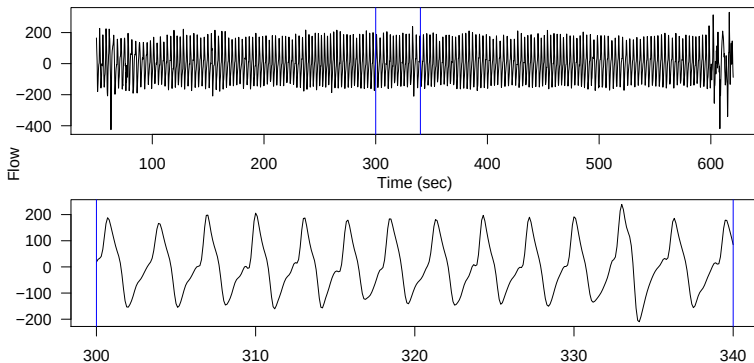
Global Annual Temperature Anomalies [Source: NASA GISS]

Surface Temperature Analysis]



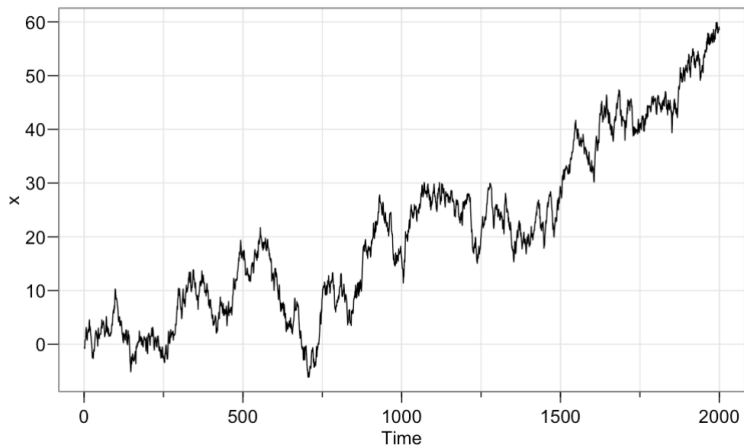
Sleep Airflow Signal

A “normal” patient’s 10 Hz sleep airflow signal [Source: [H. et al. 2022](#)]



A Simulated Time Series

```
## {r}  
set.seed(123)  
w <- rnorm(2000); x <- cumsum(w); tsplot(x, las = 1)
```



Time Series Data

- ▶ A **time series** is a collection of observations $\{y_t, t \in T\}$ taken sequentially in time (t) with the index set T
 - ▶ $T = \{0, 1, 2, \dots, T\} \subset \mathbb{Z} \Rightarrow$ **discrete-time time series**
 - ▶ $T = [0, T] \subset \mathbb{R} \Rightarrow$ **continuous-time time series**
- ▶ A discrete-time time series might be intrinsically discrete or might arise from an underlying continuous-time time series via
 - ▶ sampling (e.g., instantaneous wind speed)
 - ▶ aggregation (e.g., daily accumulated precipitation amount)
 - ▶ extrema (e.g., daily maximum temperature)
- ▶ We will focus on **discrete-time real-valued** ($Y_t \in \mathbb{R}$) **time series**

Exploratory Time Series Analysis

- ▶ Start with a **time series plot**, i.e., to plot y_t versus t 
- ▶ Look for:
 - ▶ Are there abrupt changes?
 - ▶ Are there “outliers”?
 - ▶ Is there a need to transform the data?
- ▶ Examine the **trend**, **seasonal components**, and the “noise” term

Features of Time Series

► Trends

- A trend, μ_t , represents smooth, long-term changes, typically in the mean $\Rightarrow$ *“the essential idea of trend is that it shall be smooth”* - [Kendall, 1973]
- Usually the form of the trend is unknown and needs to be estimated. When the trend is removed, we obtain a **detrended** series

► Seasonal or periodic components

- A seasonal component s_t repeats itself in time, i.e., $s_t = s_{t+kd}$
- We need to estimate the form and/or the period d of the seasonal component to **deseasonalize** the series

► The “noise” process

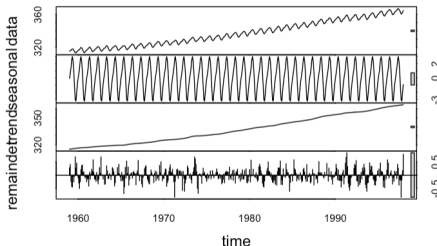
- The noise process, η_t , is the component that is neither trend nor seasonality
- We will focus on finding plausible (typically stationary) statistical models for this process

Combining Trend, Seasonality, and Noise Together

Two common decompositions are:

► Additive model:

$$y_t = \mu_t + s_t + \eta_t, \quad t = 1, \dots, T$$



► Multiplicative model:

$$y_t = \mu_t s_t \eta_t, \quad t = 1, \dots, T$$

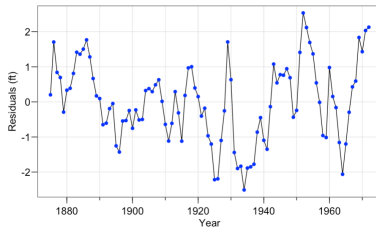
If all $\{y_t\}$ are positive then we obtain the additive model by taking logarithms:

$$\log y_t = \log \mu_t + \log s_t + \log \eta_t, \quad t = 1, \dots, T$$

Time Series Models

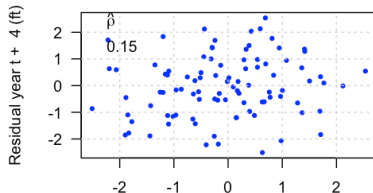
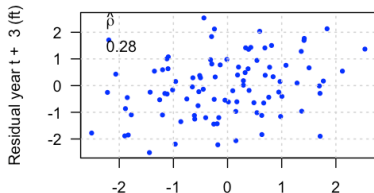
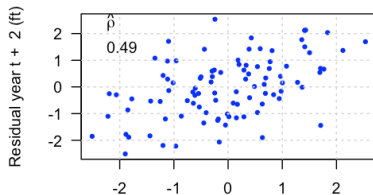
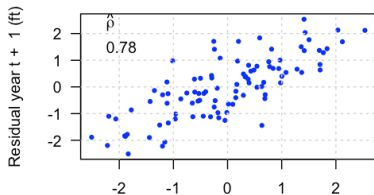
Lake Huron Time Series

- ▶ **Time series analysis** studies observations that are **dependent** across time (typically $\{\eta_t\}$)
- ▶ Some key features of the Lake Huron time series:
 - ▶ decreasing trend
 - ▶ some “random” fluctuations around the decreasing trend
- ▶ We extract the “noise” component by assuming a linear trend



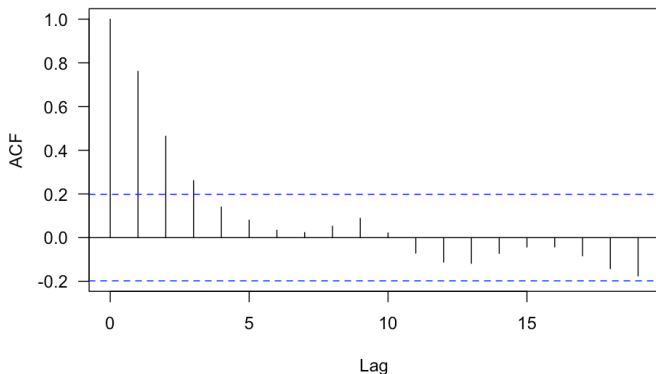
Exploring the Temporal Dependence Structure of $\{\eta_t\}$

Nearby observations tend to be more similar than observations farther apart in time. We can explore this using lag plots



Further Exploration of the Temporal Dependence Structure

Let's plot the correlation as a function of the time lag



In a few weeks we will learn how to use this information to suggest an appropriate model

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- ▶ A central simplifying assumption in time-series modeling is **stationarity** $\Rightarrow$ distributional properties depend only on time lags, not on specific time points
- ▶ Most time series are not stationary; we remove or model non-stationarity (e.g., detrending, deseasonalizing) to obtain a stationary component $\{\eta_t\}$. Typically, we assume **second-order stationarity**:

$$\begin{aligned} \mathbb{E}[\eta_t] &= 0 \text{ for all } t, \text{ and} \\ \text{Cov}(\eta_t, \eta_{t'}) &= \gamma(t' - t) = \text{Cov}(\eta_{t+s}, \eta_{t'+s}) \end{aligned}$$

Objectives of Time Series Analysis

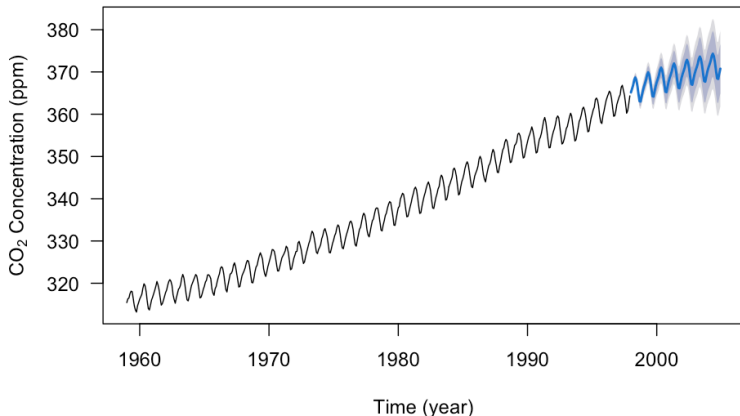
Objectives of Time Series Analysis: Modeling

- ▶ **Modeling:** find a **statistical model** that adequately explains the observed time series
- ▶ For example, identify a model that accounts for the fact that Lake Huron depths are correlated across different years and exhibit a decreasing long-term trend
- ▶ The fitted model can be used for further **statistical inference**, for instance, to answer questions such as: **Is there evidence of a decreasing trend in Lake Huron depths?**

Objectives of Time Series Analysis: Forecasting

Forecasting is perhaps the most common objective. One observes a time series of given length and wishes to **predict** or **forecast** future values based on the observed data

Forecasts from TBATS(1, {3,1}, -, {<12,5>})



Other Objectives of Time Series Analysis

- ▶ **Adjustment:** e.g., [seasonal adjustment](#), where the seasonal component is estimated and removed to better reveal the underlying trend
- ▶ **Simulation:** use a time series model (that adequately describes a physical process) as a surrogate to *simulate repeatedly and approximate the process's behavior*
- ▶ **Control:** adjust [input \(control\)](#) parameters to make the series conform more closely to a given standard (many examples come from statistical quality control)

Time Series in the Era of AI

- ▶ Time-series forecasting has long been a central problem in statistics:

past observations $\longrightarrow$ future prediction.

- ▶ Classical statistical approaches:
 - ▶ ARIMA and exponential smoothing
 - ▶ State-space models
 - ▶ GARCH and other dynamic models
- ▶ Modern ML/AI approaches:
 - ▶ Tree-based and ensemble methods
 - ▶ Recurrent neural networks (RNN/LSTM)
 - ▶ Transformers
 - ▶ Time-series foundation models

A common goal

Learn temporal structure from observed data to **understand, predict, and quantify future behavior**

What Does AI Change?

Traditional Time-Series Models

- ▶ Explicit model structure
- ▶ Carefully specified dependence
- ▶ Relatively few parameters
- ▶ Statistical interpretability
- ▶ Uncertainty quantification

Modern ML / AI Models

- ▶ Flexible nonlinear structure
- ▶ Learn complex temporal patterns
- ▶ Potentially many parameters
- ▶ Benefit from large datasets

Not “Statistics vs. AI”

Modern forecasting increasingly combines **statistical structure, machine learning, and domain knowledge**

A New Direction: Pretrained Time-Series Models

Classical statistical modeling

target dataset $\longrightarrow$ specify and fit a model $\longrightarrow$ prediction

- ▶ ARIMA: model temporal dependence within a series
- ▶ Spatio-temporal models: jointly model related series

Foundation-model paradigm

many datasets $\longrightarrow$ pretrain once $\longrightarrow$ apply to new series

- ▶ Learn transferable temporal representations across many series
- ▶ Potentially forecast unseen series without refitting (**zero-shot forecasting**)

What is different?

The key idea is not simply using many time series—classical multivariate and spatio-temporal models already do that. The distinction is **pretraining and transfer to new series**

Why Classical Time Series Still Matters

- ▶ **Dependence:** What does the past tell us about the future?
- ▶ **Uncertainty:** How much can we trust a forecast?
- ▶ **Diagnostics:** What structure has the model missed?
- ▶ **Validation:** Will performance generalize to the future?
- ▶ **Nonstationarity:** What happens when the process changes?

Key message

AI changes the models—not the need to understand **dependence, uncertainty, and validation**